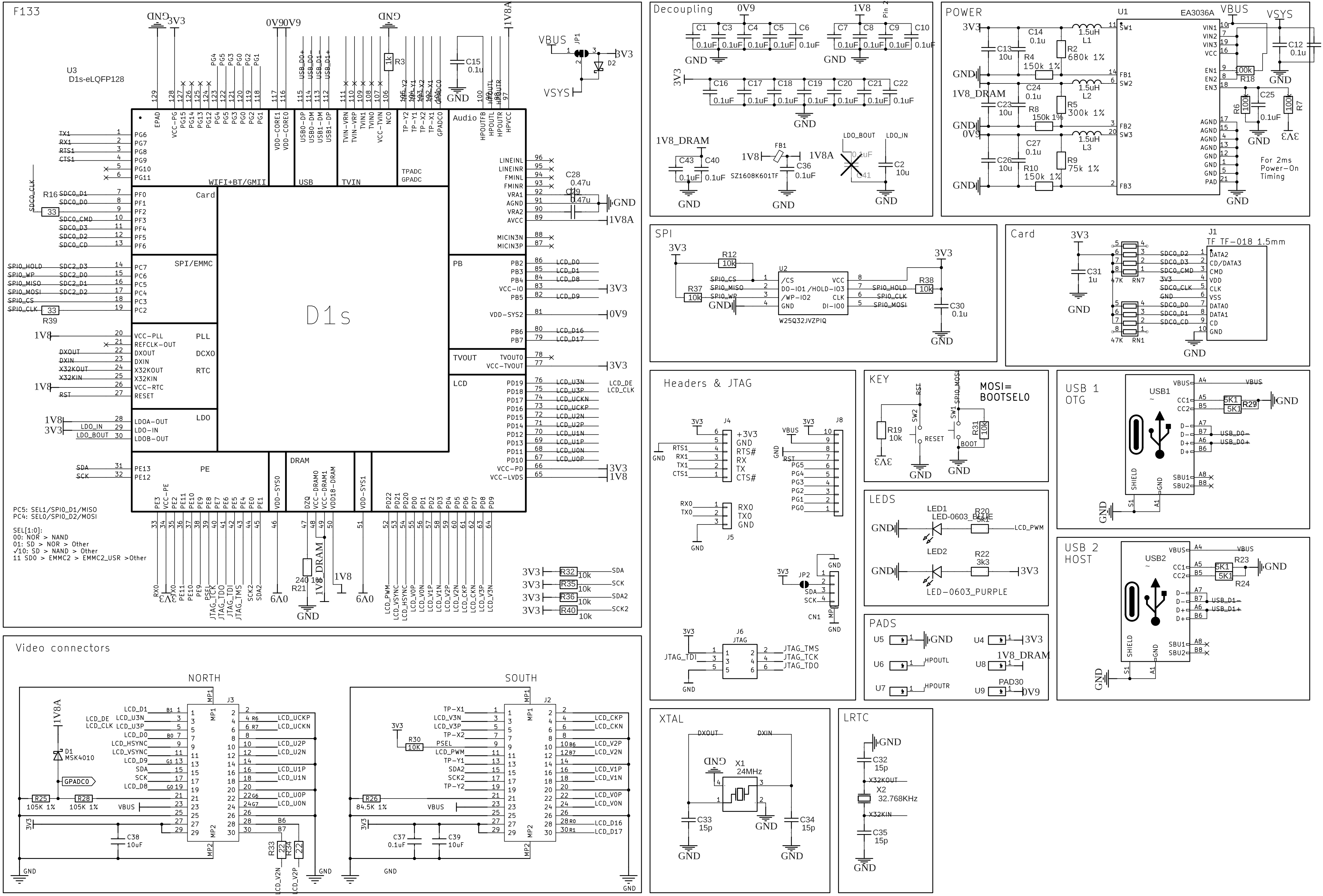
F133
U3
D1s-eLQFP128
D1s
GND
3V3
0V9
0V9
1V8A
R3
1k
C15
0.1u
VBUS
JP1
3V3
D2
VSYS
WIFI+BT/GMII
USB
TVIN
TPADC
GPADC
Audio
Card
SPI/EMMC
PLL
DCXO
RTC
LDO
PE
DRAM
PB
TVOUT
LCD
TX1
RX1
RTS1
CTS1
PG6
PG7
PG8
PG9
PG10
PG11
SDC0_CLK
R16
33
SDC0_D1
SDC0_D0
SDC0_CMD
SDC0_D3
SDC0_D2
SDC0_CD
PF0
PF1
PF2
PF3
PF4
PF5
PF6
SPI0_HOLD
SPI0_WP
SPI0_MISO
SPI0_MOSI
SPI0_CS
SPI0_CLK
SDC2_D3
SDC2_D0
SDC2_D1
SDC2_D2
R39
PC7
PC6
PC5
PC4
PC3
PC2
1V8
VCC-PLL
REFCLK-OUT
DXOUT
DXIN
X32KOUT
X32KIN
VCC-RTC
RESET
RST
LDOA-OUT
LDO-IN
LDOB-OUT
LDO_IN
LDO_BOUT
SDA
SCK
PE13
PE12
EPAD
VCC-PG
PG15
PG14
PG13
PG12
PG4
PG5
PG3
PG0
PG2
PG1
VDD-CORE1
VDD-CORE0
USB0-DP
USB0-DM
USB1-DM
USB1-DP
USB_D0+
USB_D0-
USB_D1-
USB_D1+
TVIN-VRN
TVIN-VRP
TVIN1
TVIN0
VCC-TVIN
NC0
TP-Y2
TP-Y1
TP-X2
TP-X1
GPADC0
HPOUTFB
HPOUTL
HPOUTR
HPVCC
LINEINL
LINEINR
FMINL
FMINR
VRA1
AGND
VRA2
AVCC
MICIN3N
MICIN3P
C28
0.47u
C29
0.47u
PB2
PB3
PB4
VCC-IO
PB5
VDD-SYS2
PB6
PB7
LCD_D0
LCD_D1
LCD_D8
LCD_D9
LCD_D16
LCD_D17
TVOUT0
VCC-TVOUT
PD19
PD18
PD17
PD16
PD15
PD14
PD13
PD12
PD11
PD10
VCC-PD
VCC-LVDS
LCD_DE
LCD_CLK
LCD_U3N
LCD_U3P
LCD_UCKN
LCD_UCKP
LCD_U2N
LCD_U2P
LCD_U1N
LCD_U1P
LCD_U0N
LCD_U0P
PE3
VCC-PE
PE2
PE11
PE10
PE9
PE8
PE7
PE6
PE5
PE4
PE1
PE0
VDD-SYS0
DZQ
VCC-DRAM0
VCC-DRAM1
VDD18-DRAM
VDD-SYS1
PD22
PD21
PD20
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
RX0
TX0
PSEL
JTAG_TCK
JTAG_TDO
JTAG_TDI
JTAG_TMS
SCK2
SDA2
240 1%
R21
1V8_DRAM
LCD_PWM
LCD_VSYNC
LCD_HSYNC
LCD_V0P
LCD_V0N
LCD_V1P
LCD_V1N
LCD_V2N
LCD_V2P
LCD_CKP
LCD_CKN
LCD_V3P
LCD_V3N
PC5: SEL1/SPI0_D1/MISO
PC4: SEL0/SPI0_D2/MOSI
SEL[1:0]:
00: NAND
01: SD > NOR > Other
✓10: SD > NAND > Other
11 SD0 > EMMC2 > EMMC2_USR >Other
R32 10k SDA
R35 10k SCK
R36 10k SDA2
R40 10k SCK2
Decoupling
C1 C3 C4 C5 C6 0.1uF
C7 C8 C9 C10 0.1uF
Pin 20
C16 C17 C18 C19 C20 C21 C22 0.1uF
C43 C40 0.1uF
FB1
SZ1608K601TF
C36 0.1uF
C41 0.1uF
C2 10u
POWER
U1
EA3036A
VBUS
VSYS
C13 10u
C14 0.1u
R4 150k 1%
R2 680k 1%
L1 1.5uH
SW1 FB1 SW2 FB2 SW3 FB3
C24 0.1u
C23 10u
R8 150k 1%
R5 300k 1%
L2 1.5uH
C27 0.1u
C26 10u
R10 150k 1%
R9 75k 1%
L3 1.5uH
VIN1 VIN2 VIN3 VCC
EN1 EN2 EN3
R18 100k
C12 0.1u
C11 10u
C25 0.1uF
R6 100k
R7 100k
AGND PAD
For 2ms
Power-On
Timing
SPI
R12 10k
R37 10k
U2
W25Q32JVZPIQ
/CS
DO-IO1 /HOLD-IO3
/WP-IO2
GND
VCC
CLK
DI-IO0
R38 10k
C30 0.1u
Card
C31 1u
RN7 47K
RN1 47K
J1
TF TF-018 1.5mm
DATA2
CD/DATA3
CMD
VDD
CLK
VSS
DATA0
DATA1
CD
GND
Headers & JTAG
J4
+3V3
GND
RTS#
RX
TX
CTS#
J8
RX0
TX0
GND
J5
JP2
CN1
J6
JTAG
JTAG_TMS
JTAG_TCK
JTAG_TDO
JTAG_TDI
KEY
R19 10k
SW2
RESET
SW1
SPI0_MOSI
R31 10k
BOOT
MOSI=
BOOTSEL0
LEDS
LED1
LED-0603_BLUE
R20 3k3
LED2
R22 3k3
LED-0603_PURPLE
PADS
U5 U6 U7 U4 U8 U9
HPOUTL
HPOUTR
PAD30
USB 1
OTG
USB1
R29 5K1
CC1 CC2
SHIELD
SBU1 SBU2
USB 2
HOST
USB2
R23 5K1
R24 5K1
Video connectors
NORTH
J3
D1
MSK4010
GPADC0
R25 105K 1%
R28 105K 1%
C38 10uF
R33 22
R34 22
LCD_V2N
LCD_V2P
SOUTH
J2
R30 10K
R26 84.5K 1%
C37 0.1uF
C39 10uF
XTAL
DXOUT
DXIN
X1
24MHz
C33 15p
C34 15p
LRTC
C32 15p
X2
32.768KHz
C35 15p
X32KOUT
X32KIN